

# Project Design Phase-II

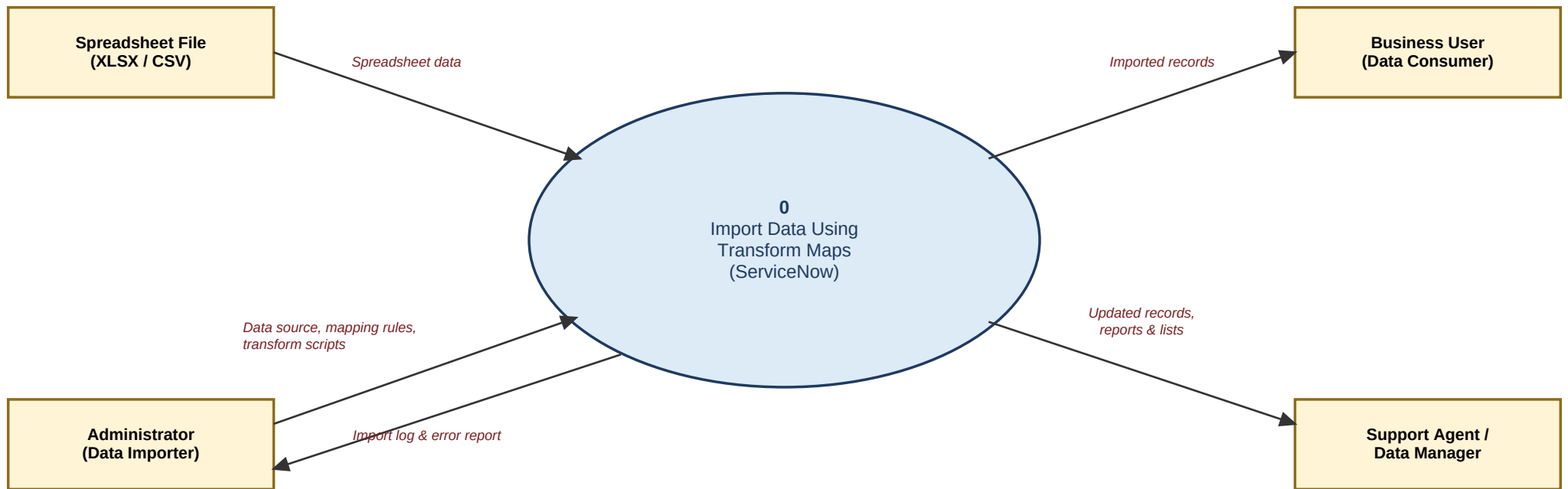
## Data Flow Diagram & User Stories

|               |                                                  |
|---------------|--------------------------------------------------|
| Date          | 03 October 2026                                  |
| Team ID       | 6ab7f10630f9767f87728ca3                         |
| Project Name  | Import Data Using Transform Maps (Spreadsheet)   |
| Maximum Marks | 4 Marks                                          |
| Team Leader   | E Dilli Kumar                                    |
| Team Members  | R Vinish, K Jerina, B Mohanavalli, P Sri Ragavan |

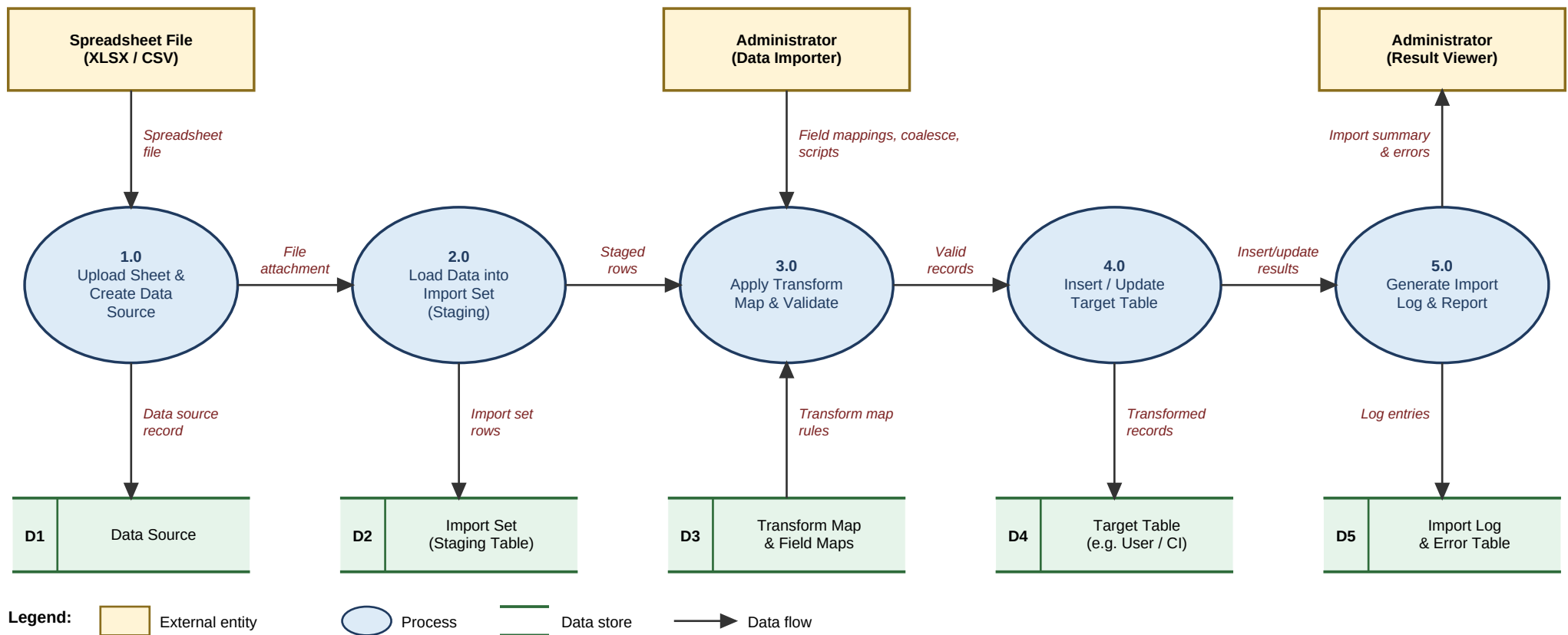
### Data Flow Diagrams

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored. For this project, the DFD describes how spreadsheet data (Excel/CSV) is imported into a ServiceNow instance: the file is uploaded to a Data Source, loaded into a staging Import Set table, converted by a Transform Map (field mapping, coalesce and transform scripts) and finally written to the target table, with every step recorded in the import log.

### Level 0 - Context Diagram



## Level 1 - Detailed Data Flow Diagram



**Flow summary:** (1) The administrator uploads the spreadsheet and creates a Data Source. (2) The file is loaded into an Import Set staging table. (3) The Transform Map applies field mappings, coalesce rules and scripts and validates each row. (4) Valid rows are inserted or updated in the target table. (5) Results and errors are written to the Import Log and shown to the administrator.

## User Stories

| User Type                        | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                                 | Acceptance criteria                                                         | Priority | Release  |
|----------------------------------|-------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------|----------|
| Administrator<br>(Data Importer) | Login & Access                | USN-1             | As an administrator, I can log into the ServiceNow instance with my credentials and open the System Import Sets module.           | I can log in and see the Import Sets menu / dashboard                       | High     | Sprint-1 |
|                                  |                               | USN-2             | As an administrator, I can be the only role (admin / import_admin) permitted to run imports.                                      | Users without the role cannot see or run import modules                     | High     | Sprint-1 |
|                                  | Data Source & File Upload     | USN-3             | As an administrator, I can create a Data Source of type File and format Excel (.xlsx) / CSV.                                      | Data Source record is saved with the chosen format                          | High     | Sprint-1 |
|                                  |                               | USN-4             | As an administrator, I can attach the spreadsheet to the Data Source and select the sheet number and header row.                  | File is attached and headers are detected correctly                         | High     | Sprint-1 |
|                                  | Load Import Set               | USN-5             | As an administrator, I can load all spreadsheet rows into a staging Import Set table.                                             | Import Set table is created and row count equals the spreadsheet rows       | High     | Sprint-1 |
|                                  |                               | USN-6             | As an administrator, I can test-load a sample (e.g. first 20 rows) before a full import.                                          | Sample rows appear in the staging table without affecting target data       | Medium   | Sprint-2 |
|                                  | Transform Map Creation        | USN-7             | As an administrator, I can create a Transform Map by selecting the source (staging) table and the target table.                   | Transform Map is saved and linked to the Import Set                         | High     | Sprint-2 |
|                                  |                               | USN-8             | As an administrator, I can use Auto Map Matching Fields to map columns that have the same name.                                   | Matching source and target fields are mapped automatically                  | Medium   | Sprint-2 |
|                                  |                               | USN-9             | As an administrator, I can manually map source fields to target fields, including reference and choice fields.                    | All required target fields are mapped and values resolve correctly          | High     | Sprint-2 |
|                                  |                               | USN-10            | As an administrator, I can set a coalesce field (e.g. email / employee ID) so existing records are updated instead of duplicated. | Re-importing the same row updates the record; no duplicate is created       | High     | Sprint-2 |
|                                  | Run Transform                 | USN-11            | As an administrator, I can run the Transform Map on the Import Set and see the number of inserted and updated records.            | Transform completes and shows insert / update / skipped counts              | High     | Sprint-2 |
|                                  |                               | USN-12            | As an administrator, I can schedule recurring imports for spreadsheets that are refreshed regularly.                              | Scheduled import runs automatically at the set time                         | Low      | Sprint-4 |
| Developer                        | Transform Scripts             | USN-13            | As a developer, I can write onBefore / onAfter and field-map scripts to clean, format or skip data during transformation.         | Script runs during transform and modified values appear in the target table | Medium   | Sprint-3 |

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|--------------------------------------|-------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------|----------|
| <b>Data Manager (Web user)</b>       | Validation & Error Handling   | USN-14            | As a data manager, I can view the Import Log and the rows that were ignored or failed, with error messages.                        | Each failed row shows the reason and can be corrected and re-imported | High     | Sprint-3 |
|                                      |                               | USN-15            | As a data manager, I will receive a notification when an import completes or fails.                                                | Email / notification is received with the import status               | Low      | Sprint-4 |
|                                      | Verify Imported Data          | USN-16            | As a data manager, I can open the target table list view and verify that imported records are correct.                             | Imported records are visible and match the spreadsheet                | High     | Sprint-3 |
|                                      |                               | USN-17            | As a data manager, I can reuse an existing Transform Map for new spreadsheets with the same format.                                | A new import uses the saved map without re-creating it                | Medium   | Sprint-3 |
|                                      | Reports                       | USN-18            | As a data manager, I can create a report or export the imported records to Excel / CSV.                                            | Report or file is generated with the imported data                    | Medium   | Sprint-4 |
| <b>Support Agent (Business User)</b> | Use Imported Data             | USN-19            | As a support agent, I can view and work with the newly imported records (e.g. users, CIs, incidents) in my normal forms and lists. | Imported records are searchable and open correctly in forms           | Medium   | Sprint-3 |
| <b>Administrator</b>                 | Admin / Audit                 | USN-20            | As an administrator, I can review the history of past imports (who imported, when, which file).                                    | Import Set history lists user, date and status                        | Low      | Sprint-4 |